A Distributional Framework for Matched Employer Employee Data

Bonhomme, Lamadon, and Manresa (2019)

Estimating wage dispersion in short panels suffers from severe bias

In short panels (e.g., 2–4 years), the number of job movers per firm is often too small to reliably estimate individual firm fixed effects.

This leads to **incidental parameter bias**:

- Overstating the variance of firm effects
- Understating (or finding negative) correlation between worker and firm effects
- Masking true sorting patterns in the data

We want to measure sorting, but standard tools force us to assume it away

Structural search and matching models (e.g., Shimer and Smith, 2000) emphasize:

- **Complementarities**: Different workers rank identical firms differently.
- **Endogenous mobility**: The decision to move depends on the current wage.
- **State dependence**: The previous firm impacts the wage at the next firm.

But empirical tractability usually forces us to use two-way fixed effects models that assume away all of these mechanisms.

The Model and Intuition

Bridging the gap

The AKM model relies on additive separability

The standard AKM (Abowd, Kramarz, and Margolis, 1999) specification:

$$Y_{it} = \alpha_i + \psi_{j(i,t)} + X_{it}'\beta + \varepsilon_{it}$$

Strong Implications:

- Log-earnings are strictly additive in worker and firm components.
- No complementarities in production or wage setting.
- Strict exogeneity: Mobility does not depend on wage realizations ε_{it}.
- No state dependence: Once you move to firm j, your previous firm j' doesn't matter.

Reducing firm heterogeneity to classes preserves parsimony

Instead of individual firm effects ψ_j, assume firms belong to K unobserved **classes**.

Let $k_{it} = k(j_{it}) \in \{1, \ldots, K\}$ be the class of the firm where worker i works at time t.

By moving the dimensionality problem from J (firms) to K (classes), we can:

- Overcome the incidental parameter bias
- Allow for complex interactions between worker types and firm classes
- Model dynamic wage paths and mobility decisions

The static model allows for non-additive interactions

Worker i has an unobserved discrete type $\alpha_i \in \{1, \ldots, L\}$.

A flexible interactive wage equation:

$$Y_{it} = a_t(k_{it}) + b_t(k_{it})\alpha_i + X_{it}'c_t + \varepsilon_{it}$$

If $b_t(k) = 1$ for all k, we recover the AKM model (at the class level). If $b_t(k)$ varies, the model captures complementarities.

The dynamic model introduces endogenous mobility and state dependence

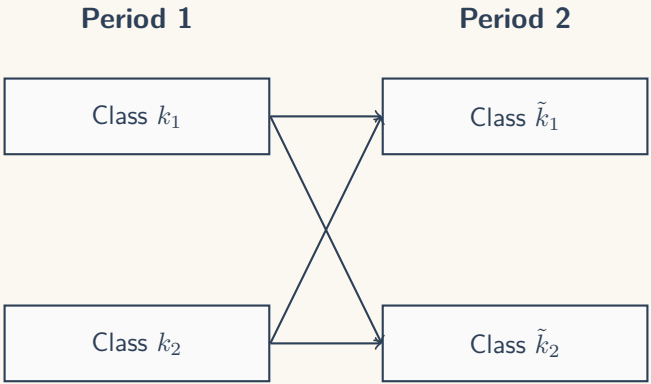
In the dynamic extension:

- **Endogenous Mobility:** The probability of moving m_{it} and the destination class k_{it+1} depend on the *current wage* Y_{it}, not just the firm class.
- **State Dependence:** The new wage Y_{it+1} depends on the *previous wage* Y_{it} and the *previous firm class* k_{it}.

This is formulated as a first-order Markov process for wages, firm classes, and mobility.

Wage functions are identified by observing workers moving across classes

How do we identify complementarities? By looking at job movers.



Identification requires asymmetric mobility patterns across classes

If log-earnings are additive, $E(Y_{i2} - Y_{i1})$ for a move from k to k' is just $a(k') - a(k)$. It doesn't depend on the mix of worker types making that move.

If there are complementarities, the wage change *does* depend on the worker type mix.

To disentangle the firm effect from the worker mix, we need a **connecting cycle** (like the previous slide) where the distribution of worker types moving $k_1 \to \tilde{k}_1$ differs from those moving $k_2 \to \tilde{k}_1$.

Step 1: Group firms using k-means clustering on empirical cdfs

First, partition the J firms into K classes by minimizing the weighted within-cluster variance of their empirical wage distributions:

$$\min_{k(1),\ldots,k(J),H_1,\ldots,H_K} \sum_{j=1}^{J} n_j \int \left(\widehat{F}_j(y) - H_{k(j)}(y)\right)^2 d\mu(y)$$

Firms with similar cross-sectional earnings distributions are grouped together.

Step 2: Estimate grouped fixed-effects using the EM algorithm

Once firms are assigned to classes k_{it}, the model becomes a standard correlated random-effects model (treating worker types α_i as the random effect).

The Sequence:

1. Estimate earnings distributions and type proportions using **job movers**.
2. Estimate type proportions for **job stayers**.
3. Combine via the EM algorithm to maximize the likelihood.

Resolution

Findings from Swedish administrative data

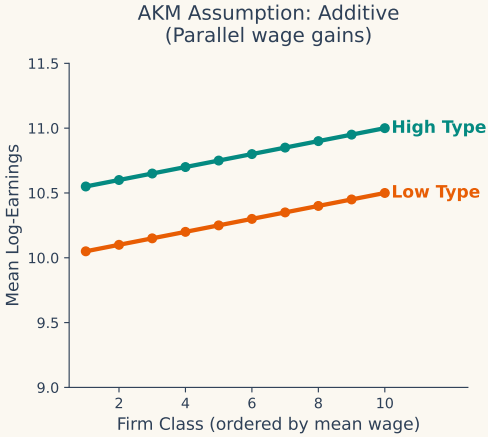
Swedish data shows additive wages but strong sorting and endogenous mobility

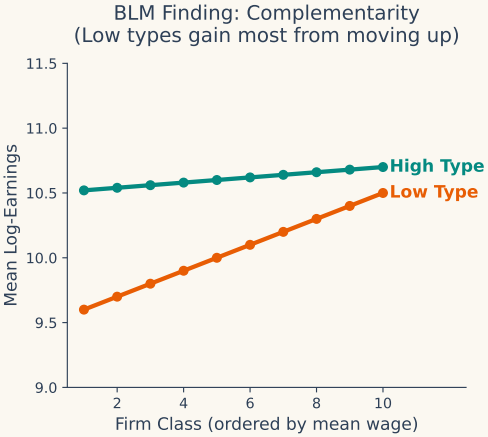
1. Variance Decomposition:
Worker heterogeneity explains $\sim 60\%$ of variance.
Firm heterogeneity (net of composition) explains $< 5\%$.

2. Strong Sorting:
The correlation between worker and firm effects is $30 - 50\%$ (much higher than typical AKM estimates).

3. Endogenous Mobility:
Workers with the lowest conditional earnings realizations (1st decile) are significantly more likely to change jobs than those in the 10th decile.

We now have a tractable bridge between reduced-form and structural models





By reducing firm heterogeneity to classes, we can tractably identify complex sorting and dynamic wage effects without the incidental parameter bias of AKM.

Appendix

Mathematical Deep Dive

Step 1: Grouping firms by minimizing distribution distances

The k-means algorithm partitions J firms into K discrete classes by minimizing:

$$\min_{k(j),H_k} \sum_{j=1}^{J} \underbrace{n_j}_{\text{Workers in firm } j} \int \left(\underbrace{\widehat{F}_j(y)}_{\text{Empirical CDF of log-wages in } j} - \underbrace{H_{k(j)}(y)}_{\text{Latent class } k \text{ representative CDF}} \right)^2 \underbrace{d\mu(y)}_{\text{Weighting measure}}$$

- This identifies firms with similar "wage signatures," regardless of the absolute level of wages.
- It reduces the dimensionality from thousands of firms to a handful of classes (e.g., $K = 10$).

The static model allows for type-specific firm premiums

For a worker i of unobserved type α_i in firm class k at time t:

$$Y_{it} = \underbrace{a_t(k_{it})}_{\text{Firm class base premium}} + \underbrace{b_t(k_{it})}_{\text{Complementarity parameter}} \cdot \underbrace{\alpha_i}_{\text{Worker ability type}} + \underbrace{X'_{it}c_t}_{\text{Time-varying observables}} + \underbrace{\varepsilon_{it}}_{\text{Idiosyncratic wage shock}}$$

- **AKM nested:** If $b_t(k) = 1$, we have additive separability.
- **Sorting:** Captured by the correlation $Corr(\alpha_i, k_{it})$.
- **Complementarities:** Captured by variation in $b_t(k)$ across classes.

The dynamic model formalizes endogenous mobility

The probability of a job move M_{it} depends on the current state:

$$\underbrace{\Pr(M_{it} = 1 \mid Y_{it}, \alpha_i, k_{it})}_{\text{Mobility Probability}} = \underbrace{\Lambda}_{\text{Logit}} \left(\underbrace{\phi(Y_{it}, \alpha_i, k_{it})}_{\text{Propensity to move}} \right)$$

And the transition to the next firm class $k_{i,t+1}$:

$$\underbrace{\Pr(k_{i,t+1} = k' \mid M_{it} = 1, Y_{it}, \alpha_i, k_{it})}_{\text{Sorting Transition}}$$

Crucially: M_{it} depends on Y_{it}. If you get a "bad" wage shock (ε_{it}), you are more likely to leave. This violates AKM's strict exogeneity.

The integrated likelihood for the dynamic model

We estimate parameters θ by maximizing the log-likelihood of the observed data:

$$\log \mathcal{L}(\theta) = \sum_{i=1}^{N} \log \sum_{\alpha=1}^{L} \underbrace{\pi(\alpha \mid k_{i1})}_{\substack{\text{Initial sorting:} \\ \text{Type dist. in } k_1}} \prod_{t=1}^{T-1} \underbrace{\mathcal{P}(Y_{it}, M_{it}, k_{i,t+1} \mid Y_{i,t-1}, \alpha, k_{it})}_{\text{Joint density of wages and mobility}}$$

- The inner sum handles the fact that worker type α is **latent** (Finite Mixture).
- The product captures the Markovian transitions over the panel.
- Solved via the **EM algorithm** for computational efficiency.